



DILIP KUMAR A N

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Objective

Engineering graduate with industry experience, seeking a challenging role to apply technical and problem-solving skills while growing in the field of electronics and embedded systems.

Experience

- **URSC/ISRO through Sree Venkateshwara Enterprises**
March-2022 - December-2023
Technical Support Engineer
 - Delivered technical support for computer systems, peripherals, and network-related issues.
 - Installed, configured, and maintained operating systems and hardware infrastructure.
 - Performed hardware diagnostics, preventive maintenance, and data recovery operations.
 - Repaired and serviced electronic devices including SMPS, printers, DFMD/HHMD systems, and radio amplifiers.

Education

Course / Degree	School / University	Grade / Score	Year
BE in ECE	R V College Of Engineering, Bengaluru	7.86	2026
Diploma in ECE	Govt. Polytechnic, Immadihalli	83%	2021
PUC in Science	Sahyadri PU College, Kolar	77.6%	2019
SSLC	Green Valley Public School	90.26%	2017

Projects

- **Solar Enerlytics: A Grid Integrated PV System with Weather Forecast Analysis**
 - Designed and developed an IoT-enabled solar energy management system using ESP32 for real-time monitoring and automated control of solar, battery, household, and grid power flow.
 - Integrated weather forecast analysis and ML-based decision-making to optimize energy utilization, battery charging, power backup, and grid energy export.
 - Built a full-stack web platform and mobile application with role-based access, real-time analytics, historical tracking, billing management, and Supabase cloud integration.
- **Power Optimization of Microprocessor using RISC-VDesign with Vedic Mathematics**
 - Designed and implemented a RISC-V based processor with a 5-stage pipeline, ALU, and MAC unit using Xilinx Vivado.
 - Applied architectural optimizations to improve performance efficiency while reducing overall power consumption.
- **Precision 6-DOF Robotic Arm for Planetary Exploration and Object Handling**
 - Designed and fabricated a 6-DOF robotic arm using custom 3D-printed components for precise object manipulation.
 - Developed embedded control mechanisms enabling smooth, low-latency motion and accurate real-time robotic operation
- **NAMMA RAITHA: Hyper-Local Agri Marketplace**
 - Developed a full-stack e-commerce platform connecting farmers and buyers with real-time inventory,

order management, and product discovery.

- Implemented role-based dashboards for farmers, buyers, and administrators using Next.js, Supabase, and Vercel.

Skills

- Hardware Description Languages (HDLs): Verilog, System Verilog.
- Hardware Platforms & Microcontrollers: FPGA Boards, Raspberry Pi, Arduino Uno, ESP8266/ESP32.
- Core ECE Concepts: Digital Logic Design, Analog Electronics, Embedded System Design, Microprocessors & Microcontrollers, and VLSI Design.
- Software & Simulation Tools: MATLAB, Simulink, Vivado, Cadence Virtuoso, Keil µVision, Arduino IDE, Pythonanywhere, Flask, Expo, Vercel.
- Programming Languages: C, Python, HTML, ReactJS
- Operating Systems: Windows, Linux (Ubuntu).

Beyond Academics

- **Beyond academics**
 - Aerospace Technologies
 - Formula 1 and Motorsport Engineering
 - Astronomy & Space Exploration
 - Emerging Engineering Innovations
 - Scientific Research and Technology Trends

Personal Details

- Date of Birth : 13/02/2002
- GitHub : github.com/dilipgowdaan